



NATIONAL UNIVERSITY
of Computer & Emerging Sciences, Lahore

Department of Computer Science

CS-1004 – Object Oriented Programming **Spring 2025**

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Course Information

Program: BS (CS)
Credit Hours: 3 + 1 for Lab
Type: Core
Class Venue:
Pre-requisites: Programming Fundamentals

Course Description/Objectives/Goals:

The core objectives of this course are to introduce,

- Object oriented programming with data abstraction and encapsulation.
- The classes, objects and relationship among different objects and classes in C++?
- Generic programming using templates, and template specializations.

Course Learning Outcomes (CLOs):

At the end of the course students will be able to:	Cognitive Level	PLO Mapping
1. Demonstrate the basic concepts of OOP	3	1
2. Apply OOP concepts (Encapsulation, Inheritance, Polymorphism, Abstraction) to computing problems for the related program	3	2
3. Model an algorithmic solution for a given problem using OOP	6	3
4. Apply good programming practices	3	3

Course Textbooks:

1. C++ Programming: Program Design Including Data Structures, by D. S. Malik (8th Edition)
2. C++: How to Program? by Deitle & Deitle (9th Edition)

Additional references and books related to the course:

3. Problem Solving with C++, by Walter Savitch
4. <https://www.learncpp.com>

Course Contents Weekly and Lecture-wise Breakdown

Week	Topic	Lecture-1	Lecture-2
1	Pointers	Pointers Introduction, Pointer variables and Initialization, Address of Operator, Dereferencing Operator. Pointer Operations (Relational, Arithmetic)	Use of Constant with Pointers. Difference between a Pointer and a Reference. Passing pointers to functions by value and by reference.
2		Dynamic memory allocation using pointers and accessing dynamic memory. Dynamic Variables new and delete operators.	Dynamic 1- dimensional arrays, Create, Delete, Grow and Shrink. Example of programs using 1D dynamic allocation: e.g., mathematical sets union and intersection.
3		Memory Leak and Dangling Pointers, Dynamic 1- dimensional char arrays for strings, string operations like search, concatenation etc.	Pointers Indirection. Dynamic 2D, allocation, matrices, CStrings etc. Introduction to smart pointers
4	Object-oriented basics	Structured Programming vs Object-oriented Programming, Principles of modularization, abstraction and encapsulation.	Objects vs Class, state vs behavior, access specifiers (Public, Private), Member functions (accessors, utilities, mutators etc)
5		Constructors (default, overloaded), Function overloading.	Dynamic memory allocation and Object assignment, Parameter passing, Shallow vs Deep copy,
6	Mid Term 1		
7	Object-oriented basics	Copy constructor, move constructor, Destructors, this pointer,	Cascaded function calls, static members, inline functions and other miscellaneous issues
8	Operator overloading	Unary operators using member functions	Binary operators using member functions
9		Binary operators using non-member functions, concept of friendship,	Unary operators, Pre and post increment, subscript operator.
10	Object and Class relationships	Part-whole relationships, Association/Aggregation	Composition Implementation issues (constructor call sequence, initializer list, etc)
11		Inheritance basics, Type of Inheritance, public, protected, private.	Function Overriding and sub-typing details
12	Mid Term 2		
13	Object and Class relationships	Polymorphism introduction Static vs dynamic binding details, virtual tables and virtual pointers,	Polymorphism vs down casting, run-time type identification, dynamic cast
14		Pure-virtual functions, Abstract classes, Interfaces (optional)	Multiple Inheritance and Diamond Problem Multiplicity, Memory Management Bi-directional relationships, Forward-class declarations issues
15	Generic Programming & Exception Handling.	Template functions	Template classes Template Specializations,
16		Exception Handling.	

(Tentative) Grading Criteria:

1. Assignments + Home works + Project **(15 %)**
 2. Quizzes **(10 %)**
 3. Midterms **(30 %)**
 4. Final Exam **(45 %)**
- Grading scheme for this course is **Absolute** under application of CS department's grading policies.
 - Minimum requirement to pass this course is to obtain at least **50%** absolute marks

Course Policies:

- All assignments and homework must be done individually.
- Late Submissions of assignments will not be accepted.
- No retake of announced quizzes
- **Plagiarism** in any work (Quiz, Assignment, Midterms, Project and Final Exam) from any source, Internet or a Student will result in **deduction of absolute marks or F** grade.
- Minimum **80%** attendance is required for appearing in the Final exams.